SCPI Command Summary

Subsystem Commands

NOTE Some [optional] commands have been included for clarity. All settings commands have a corresponding query. Not all commands apply to all models.

SCPI Command	Description
ABORt	
:ACQuire (@chanlist)	Resets the measurement trigger system to the Idle state
:DLOG	Stops a running data log (only on N6705A)
:TRANsient (@chanlist)	Resets the transient trigger system to the Idle state
CALibrate	
:CURRent	
[:LEVel] <NRf>, (@channel)	Calibrates the output current programming
:MEASure <NRf>, (@channel)	Calibrates the current measurement
:PEAK (@channel)	Calibrates the peak current limit (only on N675xA/N676xA)
:DATA <NRf>	Enters the calibration value
:DATE <SPD>, (@channel)	Sets the calibration date
:DPRog (@channel)	Calibrates the current downprogrammer
:LEVel P1 | P2 | P3	Advances to the next calibration step
:PASSword <NRf>	Sets the numeric calibration password
:SAVE	Saves the new cal constants in non-volatile memory
:STATE <Bool> [,<NRf>]	Enables/disables calibration mode
:VOLTage	
[:LEVel] <NRf>, (@channel)	Calibrates the output voltage programming
:CMRR (@channel)	Calibrates common mode rejection ratio (only N675xA/N676xA)
:MEASure <NRf>, (@channel)	Calibrates the voltage measurement
DISPlay	
[:WINDow]:VIEW METER1 | METER4	Selects 1-channel or 4-channel meter view
FETCh	(FETCh commands only on N6761A/62A and Opt. 054)
[:SCALar]	
:CURRent [:DC]? (@chanlist)	Returns the average output current
:VOLTage [:DC]? (@chanlist)	Returns the average output voltage
:ARRay	
:CURRent [:DC]? (@chanlist)	Returns the instantaneous output current
:VOLTage [:DC]? (@chanlist)	Returns the instantaneous output voltage
HCOPy	(HCOPy commands only on Agilent N6705A)
:SDUMp:DATA?	Returns an image of the display in .gif format
INITiate	
[:IMMediate]	
:ACQuire (@chanlist)	Enables measurement triggers (only N6761A/62A and Opt. 054)
:DLOG "filename"	Enables the data logger function (only on N6705A)
:TRANsient (@chanlist)	Enables output triggers
:CONTinuous	
:TRANsient <Bool>, (@chanlist)	Enables/disables continuous transient triggers

SCPI Command	Description
MEASure	
[:SCALar]	
:CURRent [:DC]? (@chanlist)	Takes a measurement; returns the average output current
:VOLTage [:DC]? (@chanlist)	Takes a measurement; returns the average output voltage
:ARRay	(ARRay commands only on N6761A/62A and Opt. 054)
:CURRent [:DC]? (@chanlist)	Takes a measurement; returns the instantaneous output current
:VOLTage [:DC]? (@chanlist)	Takes a measurement; returns the instantaneous output voltage
MMEMory	(MMEMory commands only on N6705A)
:ATTRibute? "object", "attribute"	Gets the attributes of a file system object
:DATA[:DEFinite]? "filename"	Gets the file contents; response is a definite length binary block
:DELete "filename"	Deletes a file
:EXPort:DLOG "filename"	Exports a data log from the display to a file
OUTPut	
[:STATe] <Bool> [,NORelay], (@chanlist)	Enables/disables the specified output channel(s)
:COUPle	
[:STATe] <Bool>	Enables/disables channel coupling for output synchronization
:CHANNel [<NR1> {,<NR1>}]	Selects which channels are coupled
:DOFFset <NRf>	Specifies a maximum delay offset to synchronize output changes
:MODE AUTO | MANual	Specifies the output delay coupling mode (only on N6705A)
:MAX:DOFFset?	Returns the maximum delay offset required for a mainframe
:DELay	
:FALL <NRf+>, (@chanlist)	Sets the output turn-off sequence delay
:RISE <NRf+>, (@chanlist)	Sets the output turn-on sequence delay
:PMODe VOLTage | CURRent, (@chanlist)	Sets the mode for turn on/off transitions (only on N6761A/62A)
:INHibit:MODE LATChing | LIVE | OFF	Sets the remote inhibit input
:PON:STATe RST | RCL0	Programs the power-on state
:PROTection	
:CLEar (@chanlist)	Resets latched protection
:COUPle <Bool>	Enables/disables channel coupling for protection faults
:DELay <NRf+>, (@chanlist)	Sets the over-current protection programming delay
:RELay:POLarity NORMal | REVerse, (@chanlist)	Sets the output relay polarity (only on Opt. 760)
SENSe	
:CURRent	
[:DC]:RANGe [:UPPer] <NRf+>, (@chanlist)	Selects the current measurement range (only on N6761A/62A)
CCOMpensate <Bool>, (@chanlist)	Enables/disables the capacitive current compensation
:DLOG	(DLOG commands only on N6705A)
:FUNCtion	
:CURRent <Bool>, (@chanlist)	Enables/disables current data logging
:MINMax <Bool>	Enables/disables min/max data logging
:VOLTage <Bool>, (@chanlist)	Enables/disables voltage data logging
:OFFSet <NR1>	Sets trigger offset as a percent from start of data log duration
:TIME <NRf+>	Sets the duration of the data log in seconds
:TINTerval <NRf+>	Sets the time interval between data log samples
:FUNCtion "VOLTage" | "CURRent", (@chanlist)	Selects the measurement function
:SWEep	(SWEep commands only on N6761A/62A and Opt. 054)
:OFFSet:POINts <NRf+>, (@chanlist)	Defines the trigger offset in the measurement sweep
:POINts <NRf+>, (@chanlist)	Defines the number of data points in the measurement
:TINTerval <NRf+>, (@chanlist)	Sets the measurement sample interval
:VOLTage[:DC]:RANGe [:UPPer] <NRf+>, (@chanlist)	Selects the voltage measurement range (only on N6761A/62A)
:WINDow [:TYPE] HANNing | RECTangular, (@chanlist)	Selects the window type (only on N6761A/62A and Opt. 054)

<table>
<tr><th>SCPI Command</th><th>Description</th></tr>
<tr><td>[SOURce:]</td><td></td></tr>
<tr><td>ARB</td><td>(ARB commands only on N6705A)</td></tr>
<tr><td>:COUNt <NRf+> | INFinity, (@chanlist)</td><td>Sets the Arb repeat count</td></tr>
<tr><td>:CURRent</td><td></td></tr>
<tr><td>:UDEFined</td><td></td></tr>
<tr><td>:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)</td><td>Generate triggers at the Beginning Of STep</td></tr>
<tr><td>:POINts? (@chanlist)</td><td>Returns the number of BOST points</td></tr>
<tr><td>:DWELl <NRf> {,<NRf>}, (@chanlist)</td><td>Sets the user-defined dwell values</td></tr>
<tr><td>:POINts? (@chanlist)</td><td>Returns the number of dwell points</td></tr>
<tr><td>:LEVel <NRf> {,<NRf>}, (@chanlist)</td><td>Sets the user-defined current values</td></tr>
<tr><td>:POINts? (@chanlist)</td><td>Returns the number of current points</td></tr>
<tr><td>:FUNCtion STEP | RAMP | STAircase | SINusoid | PULSe | TRAPezoid | EXPonential | UDVoltage | UDCurrent | NONE, (@chanlist)</td><td>Selects the ARB function</td></tr>
<tr><td>:TERMinate:LAST <Bool>, (@chanlist)</td><td>Sets the ARB termination mode</td></tr>
<tr><td>:VOLTage</td><td></td></tr>
<tr><td>:CONVert (@channel)</td><td>Converts the selected ARB to a user-defined list</td></tr>
<tr><td>:EXPonential</td><td></td></tr>
<tr><td>:END[:LEVel] < NRf+>, (@channel)</td><td>Sets the end voltage of the exponential ARB</td></tr>
<tr><td>:STARt</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the initial voltage of the exponential ARB</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the start time or delay</td></tr>
<tr><td>:TCONstant < NRf+>, (@channel)</td><td>Sets the time constant of the exponential ARB</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the time of the exponential ARB</td></tr>
<tr><td>:PULSe</td><td></td></tr>
<tr><td>:END:TIMe < NRf+>, (@channel)</td><td>Sets the length of the end time</td></tr>
<tr><td>:STARt</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the initial voltage of the pulse</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the start time or delay</td></tr>
<tr><td>:TOP</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the top level voltage of the pulse</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the pulse</td></tr>
<tr><td>:RAMP</td><td></td></tr>
<tr><td>:END</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets end voltage of the ramp</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the end time</td></tr>
<tr><td>:RTIMe < NRf+>, (@channel)</td><td>Sets the rise time of the ramp</td></tr>
<tr><td>:STARt</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the initial voltage of the ramp</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the start time or delay</td></tr>
<tr><td>:SINusoid</td><td></td></tr>
<tr><td>:AMPLitude < NRf+>, (@channel)</td><td>Sets the amplitude of the sine wave</td></tr>
<tr><td>:FREQuency < NRf+>, (@channel)</td><td>Sets the frequency of the sine wave</td></tr>
<tr><td>:OFFSet < NRf+>, (@channel)</td><td>Sets the DC offset of the sine wave</td></tr>
<tr><td>:STAircase</td><td></td></tr>
<tr><td>:END</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the end voltage of the staircase</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the end time</td></tr>
<tr><td>:NSTeps < NRf+>, (@channel)</td><td>Sets the number of steps in the staircase</td></tr>
<tr><td>:STARt</td><td></td></tr>
<tr><td>[:LEVel] < NRf+>, (@channel)</td><td>Sets the initial voltage of the staircase</td></tr>
<tr><td>:TIMe < NRf+>, (@channel)</td><td>Sets the length of the start time or delay</td></tr>
<tr><td>:TIMe <NRf+>, (@channel)</td><td>Sets the length of the staircase</td></tr>
</table>

SCPI Command	Description
[SOURce:]ARB continued	
:STEP	
:END[:LEVel] < NRf+>, (@channel)	Sets the end voltage of the step
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the step
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:TRAPezoid	
:END:TIMe < NRf+>, (@channel)	Sets the length of the end time
:FTIMe < NRf+>, (@channel)	Sets the length of the fall time
:RTIMe < NRf+>, (@channel)	Sets the length of the rise time
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the trapezoid
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:TOP	
[:LEVel] < NRf+>, (@channel)	Sets the top level voltage of the trapezoid
:TIMe < NRf+>, (@channel)	Sets the length of the top of the trapezoid
:UDEFined	
:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the Beginning Of Step
:POINts? (@chanlist)	Returns the number of BOST points
:DWELl <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined dwell values
:POINts? (@chanlist)	Returns the number of dwell points
:LEVel <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined voltage values
:POINts? (@chanlist)	Returns the number of voltage points
CURRent	
[:LEVel]	
[:IMMediate][:AMPLitude] <NRf+>, (@chanlist)	Sets the output current
:TRIGgered [:AMPLitude] <NRf+>, (@chanlist)	Sets the triggered output current
:MODE FIXed | STEP | LIST | ARB, (@chanlist)	Sets the current trigger mode
:PROTection	
:DELay[:TIME]<NRf+> (@chanlist)	Sets the over-current protection programming delay
:STARt SCHange | CCTRans, (@chanlist)	Sets the over-current protection programming mode
:STATe <Bool>, (@chanlist)	Enables/disables over-current protection on the selected output
:RANGe <NRf+>, (@chanlist)	Sets the output current range (only on N6761A/62A)
DIGital	
:INPut:DATA?	Reads the state of the digital port pins
:OUTPut:DATA <NRf>	Sets the digital port
:PIN<1-7>	
:FUNCtion DIO | DINPut | TOUTput | TINPut | FAULt[1] | INHibit[2] | ONCouple | OFFCouple	Sets the selected pin's function ([1]PIN1 only; [2]PIN3 only)
:POLarity POSitive | NEGative	Sets the selected pin's polarity
LIST	(LIST commands only on N6761A/62A and Opt. 054)
:COUNt <NRf+> | INFinity, (@chanlist)	Sets the list repeat count
:CURRent [:LEVel] <NRf> {,<NRf>}, (@chanlist)	Sets the current list
:POINts? (@chanlist)	Returns the number of current list points
:DWELl <NRf> {,<NRf>}, (@chanlist)	Sets the list of dwell times
:POINts? (@chanlist)	Returns the number of dwell list points
:STEP ONCE | AUTO, (@chanlist)	Specifies how the list responds to triggers
:TERMinate:LAST <Bool>, (@chanlist)	Sets the list termination mode
:TOUTput	
:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the Beginning Of STep
:POINts? (@chanlist)	Returns the number of BOST list points
:EOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the End Of STep
:POINts? (@chanlist)	Returns the number of EOST list points

SCPI Command	Description
[SOURce:]LIST continued	
:VOLTage[:LEVel] <NRf> {,<NRf>}, (@chanlist)	Sets the voltage list
:POINts? (@chanlist)	Returns the number of voltage list points
POWer:LIMit <NRf+>, (@chanlist)	Sets the power limit on output channels
STEP:TOUTput <Bool>, (@chanlist)	Generate a trigger output on the voltage or current step
VOLTage	
[:LEVel]	
[:IMMediate][:AMPLitude] <NRf+>, (@chanlist)	Sets the output voltage
:TRIGgered [:AMPLitude] <NRf+>, (@chanlist)	Sets the triggered output voltage
:MODE FIXed | STEP | LIST | ARB, (@chanlist)	Sets the voltage trigger mode
:PROTection[:LEVel] <NRf+>, (@chanlist)	Sets the over-voltage protection level
:RANGe <NRf+>, (@chanlist)	Sets the output voltage range (only on N6761A/62A)
:SENSe:SOURce INTernal | EXTernal, (@chanlist)	Sets the remote sense relays (only on N6705A)
:SLEW[:IMMediate] <NRf+> | INFinity, (@chanlist)	Sets the output voltage slew rate
STATus	
:OPERation	
[:EVENt]? (@chanlist)	Returns the value of the operation event register
:CONDition? (@chanlist)	Returns the value of the operation condition register
:ENABle <NRf>, (@chanlist)	Enables specific bits in the Event register
:NTRansition <NRf>, (@chanlist)	Sets the Negative transition filter
:PTRansition <NRf>, (@chanlist)	Sets the Positive transition filter
:PRESet	Presets all enable and transition registers to power-on
:QUEStionable	
[:EVENt]? (@chanlist)	Returns the value of the questionable event register
:CONDition? (@chanlist)	Returns the value of the questionable condition register
:ENABle <NRf>, (@chanlist)	Enables specific bits in the Event register
:NTRansition <NRf>, (@chanlist)	Sets the Negative transition filter
:PTRansition <NRf>, (@chanlist)	Sets the Positive transition filter
SYSTem	
:CHANnel	
[:COUNt]?	Returns the number of output channels in a mainframe
:MODel? (@chanlist)	Returns the model number of the selected channel
:OPTion? (@chanlist)	Returns the option installed in the selected channel
:SERial? (@chanlist)	Returns the serial number of the selected channel
:COMMunicate	
:RLSTate LOCal | REMote | RWLock	Specifies the Remote/Local state of the instrument
:TCPip:CONTrol?	Returns the control connection port number
:DATE <yyyy>,<mm>,<dd>	Sets the date of the system clock (only on N6705A)
:ERRor?	Returns the error number and error string
:GROup	
:CATalog?	Returns the groups that have been defined
:DEFine (@chanlist)	Group multiple channels together to create a single output
:DELete <channel>	Removes the specified channel from a group
:ALL	Ungroups all channels
:PASSword:FPANel:RESet	Resets the front panel lock password to zero
:REBoot	Returns the unit to its power-on state
:TIME <hh>,<mm>,<ss>	Sets the time of the system clock (only on N6705A)
:VERSion?	Returns the SCPI version number

<table>
<tr><th>SCPI Command</th><th>Description</th></tr>
<tr><td>TRIGger</td><td></td></tr>
<tr><td>:ACQuire</td><td>(ACQuire commands only on N6761A/62A and Opt. 054)</td></tr>
<tr><td>[:IMMediate] (@chanlist)</td><td>Triggers the measurement immediately</td></tr>
<tr><td>:SOURce BUS|PIN<1-7>|TRANsient<1-4>, (@chanlist)</td><td>Sets the measurement trigger source</td></tr>
<tr><td>:ARB</td><td></td></tr>
<tr><td>:SOURce IMMediate | EXTernal | BUS</td><td>Sets the arbitrary waveform trigger source (only on N6705A)</td></tr>
<tr><td>:DLOG</td><td>(DLOG commands only on N6705A)</td></tr>
<tr><td>[:IMMediate]</td><td>Triggers the data logger immediately</td></tr>
<tr><td>:CURRent</td><td></td></tr>
<tr><td>[:LEVel] <NRf>, (@chanlist)</td><td>Sets the current trigger level of the data logger</td></tr>
<tr><td>:SLOPe POSitive | NEGative, (@chanlist)</td><td>Sets the current trigger slope of the data logger</td></tr>
<tr><td>:SOURce IMMediate | EXTernal | BUS | VOLTage<1-4>
| CURRent<1-4>| ARSKey | OOOKey</td><td>Sets the data logger trigger source</td></tr>
<tr><td>:VOLTage</td><td></td></tr>
<tr><td>[:LEVel] <NRf>, (@chanlist)</td><td>Sets the voltage trigger level of the data logger</td></tr>
<tr><td>:SLOPe POSitive | NEGative, (@chanlist)</td><td>Sets the voltage trigger slope of the data logger</td></tr>
<tr><td>:TRANsient</td><td></td></tr>
<tr><td>[:IMMediate] (@chanlist)</td><td>Triggers the output immediately</td></tr>
<tr><td>:SOURce BUS|PIN<1-7>|TRANsient<1-4>, (@chanlist)</td><td>Sets the output trigger source</td></tr>
</table>

Common Commands

Command	Description	Command	Description
*CLS	Clear status	*RST	Reset
*ESE <NRf>	Standard event status enable	*SAV <NRf>	Saves an instrument state
*ESR?	Return event status register	*SRE <NRf>	Set service request enable register
*IDN?	Return instrument identification	*STB?	Return status byte
*OPC	Enable "operation complete" bit in ESR	*TRG	Trigger
*OPT?	Return option number	*TST?	Performs self-test, then returns result
*RCL <NRf>	Recalls a saved instrument state	*WAI	Pauses additional command processing until all device commands are done
*RDT?	Return output channel descriptions		